

1

Problem

FLUENT TEXT LACKS SUPPORT

- Language models know what to say and how to present it, yet often produce unsupported factoid claims.
- Retrieval-augmented generation still may ignore, exceed or contradict retrieved documents.
- Editing for Attribution must maximize evidence support while preserving the original intent, style and structure.

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Evaluation

ATTRIBUTION x PRESERVATION

- Attribution AIS measures whether every sentence is fully supported by at least one identified source; auto-AIS uses NLI.
- Preservation combines human intent similarity with character-level Levenshtein similarity.
- The harmonic mean F1AP exposes the trade-off: deleting or rewriting everything may improve support but destroy the original.

2

Research stage

QUESTION EVERY CLAIM

- PaLM uses six demonstrations to generate comprehensive verification questions; three samples are unioned for coverage.
- For each query, web search retrieves five pages and sliding windows create candidate evidence snippets.
- A relevance model selects evidence, then at most five snippets maximize query coverage for the attribution report.

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Evidence

- On PaLM Natural Questions, human AIS rises 35.4 to 43.4 while combined preservation remains 83.1; F1AP reaches 57.0.
- On StrategyQA reasoning chains, AIS rises 24.5 to 31.5 with 84.6 preservation; on QReCC, 13.2 to 28.3 with 78.1.
- RARR preserves style and structure better than EFEC or LaMDA Research and can improve downstream NQ answer accuracy.

3

Revision stage

EDIT ONLY DISAGREEMENTS

- An agreement model compares the current passage and evidence through each question, explicitly inferring both answers.
- When they disagree, an edit model identifies the offending span and minimally rewrites it to match the evidence.
- Edits are applied iteratively and rejected if their character distance is too large, controlling unnecessary drift.

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Limits & risks

- Search may retrieve weak, conflicting or biased sources; RARR verifies support, not source trustworthiness.
- Poetry, long documents, logical coherence and deciding which claims need citations remain difficult.
- Few-shot large-model calls and web research are computationally costly, and partially attributed output must be labeled clearly.

RARR

RARR

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Retrofit attribution onto any generated passage: research evidence for its claims, revise only where text and evidence disagree, and return both a minimally changed passage and a compact source report.

attribution

web research

post-editing